3-Day Terraform Training Program – Detailed Official Curriculum

DAY 1 – Terraform Foundations and Core Workflow

Module 1: Introduction to Infrastructure as Code (IaC)

- Overview of IaC and its relevance in modern DevOps and Cloud environments
- Advantages of Terraform as a provisioning tool
- Comparison of Declarative vs. Imperative approaches
- Enterprise adoption use cases

Module 2: Terraform Architecture

- Terraform CLI, Providers, Registry, and Module Framework
- Understanding Terraform State, Locking, and Execution Plan
- Infrastructure lifecycle management concepts

Module 3: Installation and Setup

- Installation of Terraform across different operating systems
- AWS CLI configuration and authentication
- Environment variables and folder structures
- Lab: Configuring Terraform and AWS authentication

Module 4: HashiCorp Configuration Language (HCL)

- Syntax, Structure, and Core Components (Variables, Outputs, Locals)
- Conditional expressions and iteration constructs
- Dynamic blocks and advanced configuration patterns
- Lab: Creating the first Terraform configuration template

Module 5: Terraform Command Workflow

• Introduction to init, validate, plan, apply, and destroy commands

• Best practices for execution workflow

• Lab: Deploying a basic EC2 instance

DAY 2 – State Management, Provisioners, and Modules

Module 6: State Management

• Understanding terraform.tfstate and its role in infra tracking
• Remote State configuration using AWS S3
• State locking using DynamoDB
• terraform state subcommands (list, show, mv, rm)
• Lab: Implementing remote backend with state locking

Module 7: Variables, Outputs, and Data Sources

• Defining and using variables
• Sensitive input handling and tfvars files
• Working with data sources to reference existing infrastructure
• Lab: Parameterized configuration setup

Module 8: Provisioners

• Overview of local-exec and remote-exec
• Secure remote execution using SSH
• Recommended and discouraged use cases
• Lab: Configuring EC2 using provisioners

Module 9: Terraform Modules (Foundational to Advanced)

• Purpose and structure of Terraform modules
• Public and private registry modules
• Module versioning and best practices
• Lab: Using AWS VPC module, creating custom modules, multi-environment modules

Module 10: Advanced HCL Constructs
• count vs. for_each
• Handling complex objects and data structures
• CIDR calculations and reusable patterns
• Lab: Auto-generating subnets using HCL loops

DAY 3 – Advanced Terraform, CI/CD Integration, and Best Practices

Module 11: Workspaces

- Logical environment separation using workspaces
- Appropriate use cases and limitations
- Lab: Managing dev, staging, and production through workspaces

Module 12: Terraform Cloud

- Remote operations, VCS integration, and policy enforcement
- Role-based access and secure workspace management
- Lab: Executing Terraform runs from Terraform Cloud

Module 13: CI/CD Integration (Industry-Oriented)

- Integrating Terraform with GitHub Actions, Azure DevOps, and Jenkins
- Automated plan and apply pipelines
- Introduction to Terratest for infrastructure testing
- Lab: Implementing GitHub Actions pipeline

Module 14: Security and Compliance

- Secrets management and Vault integration
- Terraform security scanning using tfsec and Checkov
- IAM role-based access for Terraform automation
- Lab: Security scanning and compliance implementation

Module 15: Debugging and Performance Optimization

- Terraform debugging using TF_LOG
- Identifying and resolving dependency cycle issues
- Improving performance through optimized resource structure
- Lab: Troubleshooting failed deployments

Module 16: End-to-End Infrastructure Project

- Deployment of a complete multi-tier architecture:
  - VPC, Subnets, NAT, EC2, ALB, RDS, IAM, CloudWatch
- Integration of modules, workspaces, and remote backend
- Lab: Full infrastructure rollout and teardown